

Technical & Functional Specifications

Document: Decentralized Inbound Tourism Platform

1. Executive Summary & System Overview

1.1 Project Mission

The platform is an Online Travel Agency (OTA) tailored for authentic, community-based Cuban tourism, emphasizing ecotourism, agrotourism, rural excursions, and cultural immersion beyond traditional resort circuits.

1.2 System Ecosystem

The system comprises three decoupled yet synchronized applications:

- Client-Facing Web Application (Frontend):** A responsive, global web platform allowing international travelers to discover Cuban heritage, configure customized itineraries, book pre-assembled packages, and execute secure international payments.
- Operations Back-Office (Havana Operations ERP):** An administrative dashboard enabling dispatch management, reservation lifecycle monitoring, operational escalation, legal document generation, and worker compliance auditing.
- Worker Mobile Application (Field Operations):** A lightweight, low-bandwidth hybrid mobile application for on-the-ground service providers (guides, drivers, accommodation hosts) to manage availability calendars, accept or decline dispatched assignments, and review task itineraries.

2. User Roles & Access Control (RBAC)

Role	Access Scope	Assigned Entities	Permissions
Super Admin	Global / Unrestricted	Founders / Lead Engineers	Full database CRUD, financial configuration, API keys, role assignment, platform deployment parameters, system logs.
Operations Admin	Operational ERP	Head of Operations (Sister-in-Law)	Booking lifecycle management, manual worker

			dispatch, timeout escalation handling, direct customer communication, supplier compliance auditing.
Administrative Support	Operational ERP (Restricted)	Administrative Personnel	Worker credential review, verification document checks, reporting exports, invoice reconciliation.
Service Worker	Worker Mobile App	Guides, Drivers, Private Hosts, Translators	Availability calendar updates, job offer response (Accept/Decline), itinerary review, customer emergency contact, credential upload.
Guest / Traveler	Client Web Platform	International Visitors	Package browsing, interactive map exploration, dynamic package builder, payment submission, profile and voucher retrieval.

3. Client-Facing Web Application (Storefront)

3.1 Architecture & Performance Target

- **Target Audience:** International visitors (Europe, Canada, Latin America, worldwide).
- **Network Constraint:** Optimized for global low-latency CDNs; Core Web Vitals targets: LCP < 1.8s, FID < 100ms, CLS < 0.1.

- **Localization:** Multilingual architecture (Default: Spanish, English, French).

3.2 Visual & Discovery Engine

- **Cultural & Destination Content Hub:** Editorial modules covering cultural events, local culinary routes, historic circuits, and eco-reserves.
- **Dynamic Promotional Banner System:**
 - Displays high-priority cultural events (e.g., Havana Jazz Festival, International Ballet Festival).
 - Direct action trigger: *"Request Event Access / Ticket Concierge"*, linking event access directly into package configurations.
- **Interactive SVG Map of Cuba:**
 - Vector-rendered, province-by-province interactive map.
 - Active province states based on real-time operational availability (e.g., Havana, Matanzas, Pinar del Río, Cienfuegos, Sancti Spiritus, Ciego de Ávila).
 - Hover / Touch interactions: Highlights active nodes, surfaces brief metrics (available eco-tours, partner casas particulares, historical trails).
 - Deep-linking: Clicking a province filters the catalog to that territory's custom activities and accommodations.

3.3 Product Catalog & Dynamic Package Builder

1. **Pre-Made Curated Packages:**
 - Packaged multi-day itineraries (e.g., *"Viñales Agro-Ecology Trail"*, *"Trinidad Heritage & Mountain Immersion"*).
 - Fixed schedules with customizable accommodation tiers (Standard vs. Premium Casa Particular).
2. **Custom Package Builder (Dynamic Itinerary Engine):**
 - **Step 1 (Dates & Geographic Route):** User defines start/end dates and orders target provinces via dynamic multi-stop routing.
 - **Step 2 (Transportation Selection):**
 - Options: Independent licensed taxi, classic vintage car, minivan, or localized point-to-point transfers.
 - Vehicle preferences with assigned drivers: visual selection of vehicle classification (Sedan, Classic 1950s, Minivan) with amenity tags (A/C verification, luggage capacity).
 - **Step 3 (Lodging Configuration):** Filter verified *casas particulares* by agro-homestay, colonial heritage, or eco-villa.
 - **Step 4 (Activities & Local Specialists):** Add localized hiking guides, translation services, farm-to-table culinary experiences, and airport meet-and-greet.
 - **Step 5 (Cost Aggregation & Summary):** Transparent real-time pricing breakdown showing base transportation, lodging fees, guided activities, and applicable agency fees.

3.4 Supplier Recruitment Portal

- Public-facing registration page: */join-our-network*.
- Digital onboarding form for independent Cuban tourism professionals:
 - Category select: Tour Guide, Private Driver (*Porteador Privado*), Homestay Host (*Arrendador*), Cultural/Gastronomic Venue.

- Identity collection: Carnet de Identidad, phone/WhatsApp contact, operational municipalities.
- Credential upload field: Compulsory high-resolution capture of Formatur Credential (for guides), Transport Operating License (for drivers), or National Tourism Registry (RTN) certification.
- Submission creates a pending audit record in the Operations Back-Office.

3.5 Traveler Authentication, Messaging & Retention Lifecycle

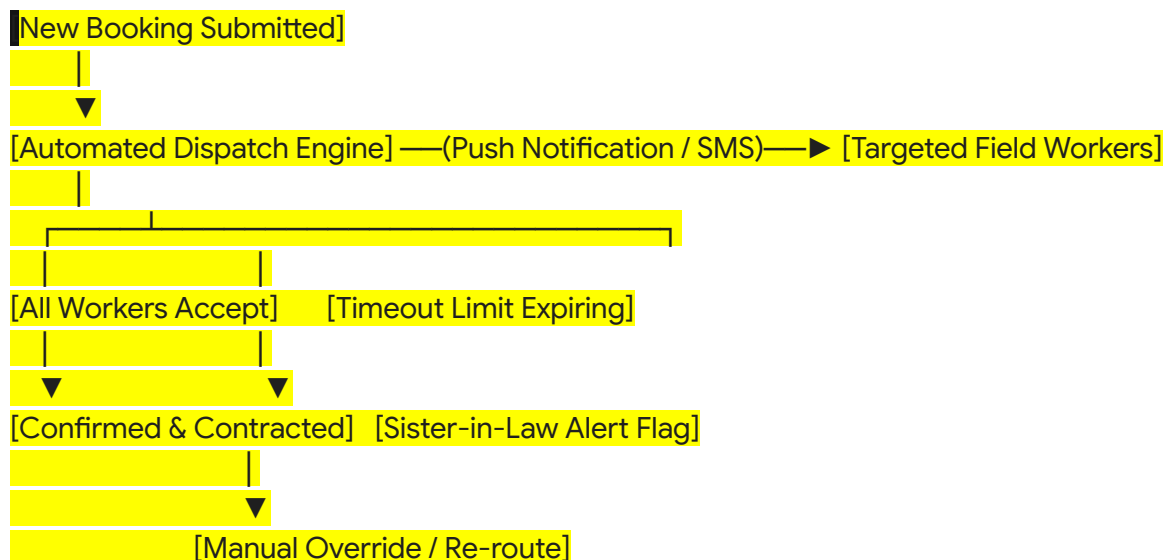
- **Lightweight Authentication:**
 - Passwordless email magic-link or single-sign-on (OAuth2: Google / Apple / Email + secure passkey).
 - Guests can browse, build itineraries, and submit requests as guests, but an account is provisioned dynamically upon booking submission (ITINERARY_SUBMITTED).
- **Self-Service Dashboard:**
 - **Visual Pipeline Tracker:** Real-time progress tracker reflecting the current reservation state machine (SUBMITTED $\rightarrow$ DISPATCHING $\rightarrow$ SECURED & INVOICED $\rightarrow$ CONFIRMED $\rightarrow$ IN PROGRESS $\rightarrow$ COMPLETED).
 - **Document Vault:** Direct download links for generated PDF proposals, invoices, payment receipts, vouchers, and emergency contact pamphlets once unlocked.
- **Integrated In-App Messaging Channel:**
 - Bi-directional communication channel between the traveler and the Havana Operations desk (Operations Admin).
 - Real-time messaging with asynchronous email fallback (if the traveler is offline, an email notification is dispatched with the operator's message and a direct portal link).
- **Automated Data Retention & GDPR Lifecycle:**
 - **Trigger Condition:** Exactly 6 months after the latest reservation transitions to COMPLETED.
 - **Retention Notification Job (Cron / Scheduled Queue):**
 - The system sends an automated transactional email notifying the user that their profile, travel history, and uploaded passport details are scheduled for deletion to protect their data privacy.
 - Call to Action: A 1-click tokenized confirmation link: [Keep My Account Active].
 - **Execution Logic:**
 - **If confirmed:** The account retention flag retention_consent_granted_at is updated to the current timestamp, extending the profile for another 12 months.
 - **If ignored (grace period: 30 days):** An automated purge worker anonymizes or soft-deletes the traveler record, scrubbing sensitive PII (passport numbers, full names, phone numbers) while preserving

anonymized aggregate metrics (financial totals, destination visit counts) for fiscal reporting.

4. Administrative Back-Office (Havana Operations ERP)

4.1 Dispatch & Reservation Tracking Matrix

The core of the back-office is an operational pipeline displaying incoming, pending, confirmed, active, and completed reservations.



4.2 Status State Machine

1. **ITINERARY_SUBMITTED**: The traveler builds their package and checks out without paying.
2. **DISPATCH_IN_PROGRESS**: The system pushes requests to the field workers' mobile apps.
3. **ASSEMBLY_&_ESCALATION**: The operations desk monitors timeouts. If a specific accommodation or guide declines, alternative workers are manually or automatically re-routed.
4. **SECURED_&_INVOICED**: All providers have accepted. The system compiles the final PDF summary, compliance contracts, and generates a secure payment link.
5. **PENDING_PAYMENT**: The client receives the invoice and has a set window (e.g., 48 hours) to complete the transaction.
6. **CONFIRMED**: Funds clear. The system automatically dispatches the final welcome

packets, pamphlets, and operational vouchers.

7. **ACTION_REQUIRED (FLAGGED)**: One or more workers have failed to confirm within the allotted timeout threshold, or a worker declined the booking.
8. **IN_PROGRESS**: Client has arrived on-site; active service delivery.
9. **COMPLETED**: Tour concluded, supplier payouts cleared for processing.
10. **CANCELLED**: Booking voided; cancellation workflow triggered per terms of service.

4.3 Automated Escalation Engine

- **Timeout Threshold**: Configurable countdown timer (e.g., 2 hours for standard bookings, 30 minutes for bookings made under 48 hours before arrival).
- **Warning States**:
 - **Amber Alert**: 75% of timer expired without worker acceptance.
 - **Red Alert**: 100% of timer expired or job explicitly declined.
- **Operations Admin Intervention**:
 - System sounds visual and audio cues on the Operations Admin dashboard.
 - Displays direct click-to-call links for the worker's primary telephone line.
 - Provides a single-click fallback re-dispatch menu displaying nearby qualified workers available for the same service, date, and location.

4.4 Document Generation Engine

Upon transition to CONFIRMED, the ERP automatically compiles and issues legal PDF documentation:

- **Traveler Voucher (Bono de Servicio)**: Detailed breakdown of itineraries, rendezvous points, emergency numbers, and the official Travel Agency License identifier.
- **Supplier Work Order (Orden de Trabajo)**: Localized assignment contract detailing payment rates, schedule, guest names, and emergency protocols.
- **Digital Invoice / Receipt**: Official itemized breakdown for traveler records.

4.5 Compliance & Worker Auditing Module

- Repository of all registered suppliers.
- Visual document inspector for uploaded Formatur credentials, driver licenses, and RTN cards.
- Expiration alert system: Flags workers whose operational or automotive licenses are within 30 days of expiration, temporarily pausing their availability for auto-dispatch.

4.6 Dynamic Inventory & Content Management System (CMS)

- **Storefront Control**: Full CRUD (Create, Read, Update, Delete) interface for all entities displayed on the client website.
- **Supplier & Asset Management**: Operations Admin can manually add, edit, disable, or verify hotels, *casas particulares*, transport providers, and tour guides.
- **Pricing Engine**: Interface to adjust seasonal rates, update agency markups, and change package details. All changes sync in real-time to the client-facing web application.

4.7 Bulk Data Import Engine (Excel/CSV)

- **Spreadsheet Ingestion:** Upload module supporting `.xlsx` and `.csv` files to seamlessly ingest pricing and package grids provided by state-owned tourist companies and hotel chains.
- **Dynamic Column Mapping:** A visual staging interface where admins match the columns from the uploaded Excel file (e.g., "Room Type", "High Season Price") to the internal database schema.
- **Validation & Commit:** The system displays a preview grid to flag data errors (e.g., missing prices) before committing the imported data to the live catalog.

4.8 AI-Powered Administrative Assistant

- **Contextual Communication:** LLM (Large Language Model) integration embedded within the messaging and email modules. The AI can instantly draft contextual, professional emails to clients (e.g., itinerary adjustments) or suppliers based on the specific reservation data.
- **Automated Reporting:** Generates automated natural-language summaries of daily operations, pending dispatch bottlenecks, and weekly financial reports.
- **Multilingual Support:** AI-assisted translation bridging communication gaps between the operations desk and global clients.

4.9 Business Intelligence, Analytics & Regulatory Reporting (KPI Dashboard)

A real-time analytical dashboard aggregating operational efficiency, financial performance, supplier quality, and mandatory fiscal/statutory reports.

4.9.1 Financial & Revenue KPIs

- **Gross Booking Value (GBV) & Net Revenue:** Real-time tracking of total transaction volume in EUR, broken down by payment rail (SEPA/Open Banking vs. Credit/Debit Card).
- **Take Rate & Margins:** Gross margin tracking per package type (Pre-assembled packages vs. Custom Dynamic Itinerary builds).
- **Average Order Value (AOV):** Mean spend per traveler and per booking transaction.
- **Supplier Payout Liabilities:** Accrued accounts payable to independent field providers (guides, drivers, hosts) vs. settled payouts, categorized by province and service classification.
- **Reinvestment Tracking:** Dedicated ledger tracking retained earnings allocated for re-investment, calculating the tax exemption threshold under Resolución 193/2026.

4.9.2 Regulatory & Tax Compliance Reporting (MINTUR & ONAT)

- **Ecotourism / Specialized Tourism Ratio:** Automatic classification and percentage breakdown of bookings qualifying as ecotourism, agrotourism, or nature trails vs. general cultural travel to support the certified 25% corporate tax rate (Resolución 193/2026).
- **Annual MINTUR Activity Summary Generator:** One-click compilation of the mandatory annual activity report required by Article 24(p) of Resolución 68/2026, summarizing tourist volume, nationalities, bed-nights generated, and geographic circuits utilized.
- **Fiscal Export Engine:** Automated export of ledger transactions, platform commission invoices, and withholding figures formatted for Cuban tax authorities (ONAT) and commercial banking reconciliations.

4.9.3 Operational & Dispatch Efficiency Metrics

- **Dispatch Latency & Acceptance Rate:** Average elapsed time between automated dispatch release and worker acceptance.
- **Timeout & Fallback Rate:** Percentage of dispatch cycles hitting the Amber/Red timeout limit requiring manual intervention by the Operations Admin.
- **Fulfillment Reliability Score:** Worker-level acceptance rates, decline rates, and last-minute cancellation frequency.
- **Booking Conversion Funnel:** Traveler journey tracking from dynamic package configuration to **ITINERARY_SUBMITTED**, **SECURED_&_INVOICED**, and **CONFIRMED**.

4.9.4 Service Quality, Experience & Duty of Care

- **Customer Satisfaction (CSAT) & Net Promoter Score (NPS):** Aggregate and granular ratings collected via the Client Mobile App post-trip.
- **Provider Scorecards:** Performance metrics per individual guide, driver, and accommodation host based on customer reviews, cleanliness ratings, punctuality, and language capability.
- **Incident & Emergency Log:** Audit trail tracking reported medical, mechanical, or logistical incidents, resolution turnaround times, and duty-of-care protocol activations.

4.9.5 Data Export & Visualizations

- **Interactive Charting:** Dynamic time-series graphs (daily, weekly, monthly, annual, seasonal tourist cycles) with geographic heat-maps of active Cuban provinces.
- **Scheduled Exports:** Automated distribution of weekly KPI executive digests via email to Super Admins (Founders) in PDF and Excel (.xlsx) formats.

5. Mobile Application Ecosystem

5.1 Technology & Low-Bandwidth Optimizations

- **Application Architecture:** Hybrid mobile clients (React Native / Flutter) sharing a unified component library.

- **Offline-First Caching:** Local SQLite / IndexedDB cache preserving critical data (schedules, client contacts) even during network disconnects.
- **Data Footprint:** Strict payload compression; JSON transfers stripped of redundant metadata to accommodate unstable 3G/4G connectivity.

5.2 Worker Mobile Application (Field Operations)

- **Authentication:** Telephone number or identity-based login with credential status banner (Active, Under Audit, Expiring Soon).
- **Availability Calendar:** Single-tap toggle for availability and granular blocking for specific days.
- **Dispatch & Job Offer:** Push notifications for dispatched jobs with countdown timers. Includes [ACCEPT] and [DECLINE] triggers with reason prompts.
- **Active Itinerary:** Offline-accessible client name, pickup location, emergency dispatch hotline, and one-touch “Report On-Site Emergency” button.

5.3 Client Mobile Application (Traveler Companion)

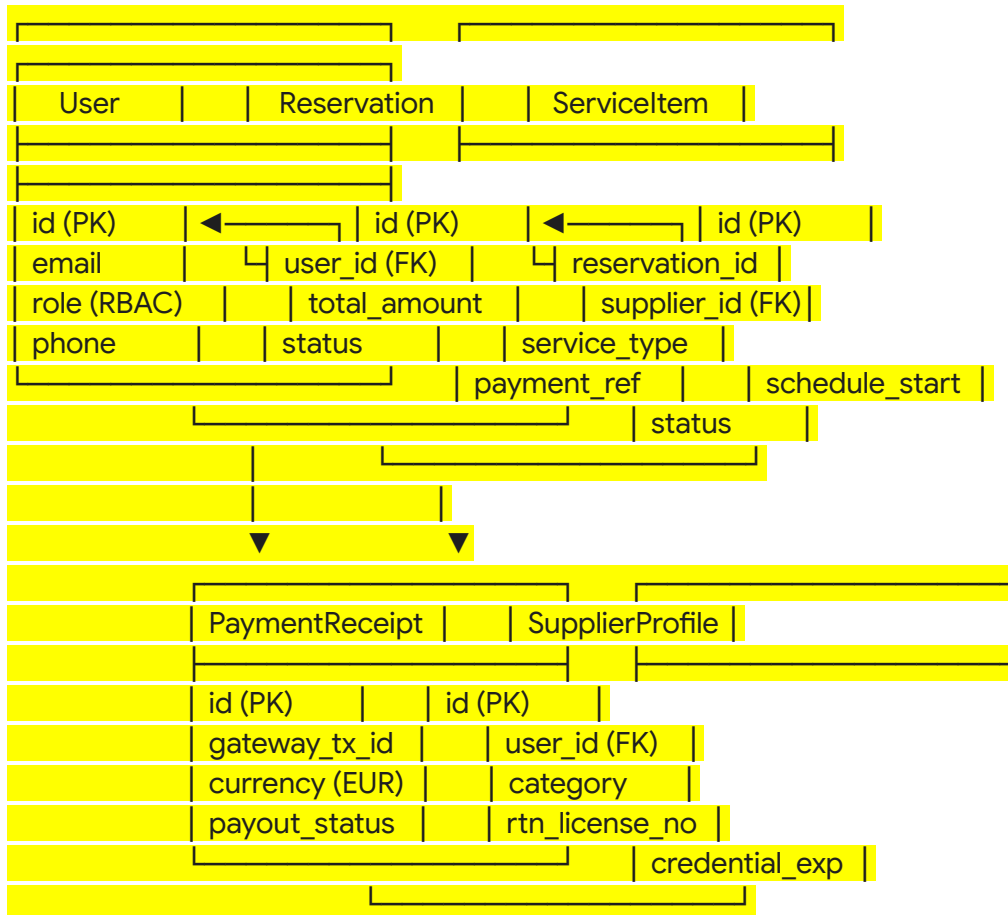
- **Self-Service Dashboard:** Access to the user profile, booking statuses, and the document vault (downloadable PDF vouchers and receipts).
- **Offline Itinerary Viewer:** Travelers can view their day-by-day schedule, rendezvous points, and assigned worker details without requiring active Wi-Fi or roaming data.
- **Push Notifications:** Real-time alerts for payment links, booking confirmations, and immediate pickup reminders.
- **Feedback & Rating Engine:** Post-service interface prompting the client to review and rate specific segments of their trip (drivers, guides, hosts).
- **In-App Messaging:** Direct chat interface connecting the traveler to the Havana Operations ERP for real-time support.

5.4 Admin Mobile Application (Operations Lite)

- **Purpose:** A lightweight triage application designed strictly for out-of-office emergency management and quick approvals, not full ERP data entry.
- **Pipeline Monitoring:** View the active reservation state machine and track live bookings.
- **Escalation Handling:** Receive push notifications for Red Alert timeouts (when a worker fails to accept a job). Admin can execute a single-click fallback re-dispatch from their phone.
- **Universal Comms:** Unified inbox allowing the Admin to reply to urgent messages from both clients and service workers while away from the desktop.

6. Data Models & Database Entities

6.1 Entity Relationship Diagram (Conceptual Layout)



6.2 Data Dictionaries

Entity: Reservations

- id: UUID (Primary Key).
- user_id: UUID (Foreign Key -> Users.id).
- booking_code: String (Unique 8-character human-readable identifier).
- created_at: Timestamp UTC.
- start_date: Date.
- end_date: Date.
- status: Enum (DRAFT, PENDING_PAYMENT, DISPATCHING, ACTION_REQUIRED, CONFIRMED, IN_PROGRESS, COMPLETED, CANCELLED).
- total_currency: String (Default: EUR).
- total_amount: Decimal (10, 2).
- custom_itinerary_payload: JSONB (Stores full dynamic package tree, client requests, and selected add-ons).

Entity: ServiceItems (Assigned Sub-Components of a Reservation)

- id: UUID (Primary Key).
- reservation_id: UUID (Foreign Key -> Reservations.id).
- supplier_id: UUID (Foreign Key -> SupplierProfiles.id, Nullable prior to assignment).
- service_type: Enum (GUIDE, TRANSPORTATION, ACCOMMODATION, EXPERIENCE).
- service_date_start: Timestamp UTC.
- service_date_end: Timestamp UTC.
- status: Enum (UNASSIGNED, OFFERED, ACCEPTED, DECLINED, TIMEOUT, FULFILLED).
- dispatch_deadline: Timestamp UTC.
- payout_rate: Decimal (10, 2).

Entity: SupplierProfiles

- id: UUID (Primary Key).
- user_id: UUID (Foreign Key -> Users.id).
- category: Enum (TOUR_GUIDE, PRIVATE_DRIVER, HOMESTAY_HOST, TRANSLATOR).
- primary_phone: String.
- provinces_active: Array [String].
- vehicle_details: JSONB (Make, Model, Year, Plate, AC_Equipped, Capacity).
- rtn_license_number: String (Unique).
- credential_document_url: String (Encrypted Object Storage URI).
- credential_expires_at: Date.
- verification_status: Enum (PENDING_AUDIT, VERIFIED, REJECTED, SUSPENDED).
- is_available: Boolean (Default: True).

7. Integration, Payments & Offline Synchronization

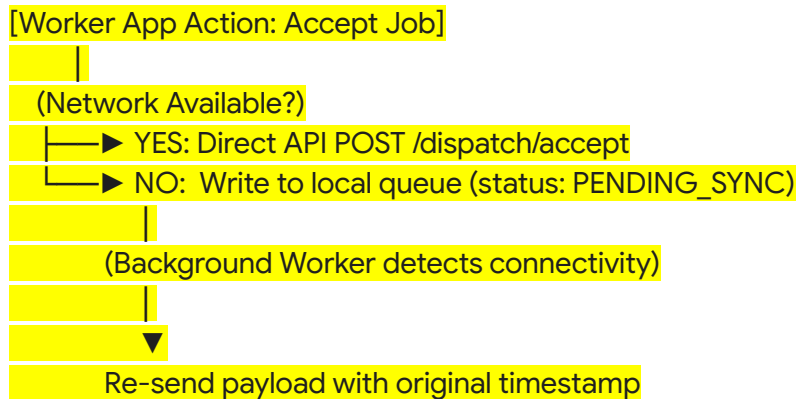
7.1 International Payment Pipeline (Revised)

- **Workflow:** Invoice-based. Payment links are generated *only after* the booking status reaches **SECURED_&_INVOICED**.
- **Checkout UI:** Multi-gateway checkout page featuring:
 - **Primary Rail:** Open Banking / SEPA Bank Transfer API (Direct account-to-account settlement via European IBAN).
 - **Secondary Rail:** Credit/Debit Card Gateway (Visa, Mastercard) to accommodate non-SEPA clients (e.g., North Americans). Includes Digital Wallet support (Apple Pay / Google Pay).
- **Currency Handling:** Base pricing is calculated in Euros (**EUR**). For non-EU clients (e.g., CAD), the card gateway will process the transaction dynamically based on the daily exchange rate.

7.2 Low-Bandwidth Synchronization Protocol (Worker App)

- **Local Data Store:** IndexedDB / SQLite database embedded within the mobile client.
- **Sync Strategy:** Two-way differential synchronizer triggered on network reconnect:
 - **Outbound Sync:** Local calendar changes, job acceptance/decline decisions queued in

- an offline operational table (sync_queue) with exponential backoff retry logic.
- **Inbound Sync:** Lightweight delta payload retrieving only state updates executed since the last known synchronization timestamp (last_synced_at).



8. Compliance, Verification & Operational Guardrails

8.1 Regulatory Invariants

- **Mandatory Provider Legality:** Under no circumstances may the system auto-dispatch an assignment to any service worker whose verification_status is not explicitly set to VERIFIED by an administrator.
- **MINTUR License Visibility:** Every generated voucher, invoice, public promotion, and booking confirmation must programmatically inject the official Travel Agency License number issued by MINTUR.
- **Duty of Care & Emergency Traceability:** The back-office must maintain a real-time tracking interface during active guest itineraries, including full emergency contact channels to local health authorities, consular services, and civil defense.

8.2 Security & Data Privacy

- **Data Protection:** Personal identity data (passport numbers, foreign billing addresses) encrypted at rest using AES-256 and transmitted exclusively via TLS 1.3.
- **Separation of Concerns:** Workers only receive traveler names, language preferences, and emergency local telephone numbers. Under no circumstances are foreign billing addresses or payment records transmitted to the Worker Mobile App.

9. Development Phases & Implementation Roadmap

(This is just an example, can be modified as needed).

Phase 1: Foundation (Weeks 1-4)

- Core database schema deployment & multi-role auth (Super Admin, Ops Admin, Worker, Guest)
- Storage bucket setup for supplier documents
- Static international storefront setup (cultural hubs, packaging layouts)

Phase 2: Booking Engine & Map Architecture (Weeks 5-8)

- Interactive SVG Cuba map module with province filtering
- Dynamic itinerary package builder with dynamic pricing calculation
- External payment gateway integration (TropiPay API checkout & webhooks)

Phase 3: Back-Office ERP & Document Automation (Weeks 9-12)

- Operations dispatch dashboard (dispatch queues, state transitions)
- Escalation engine with automated visual/audio countdown alerts
- PDF generation pipeline (travel vouchers, work orders, legal receipts)
- Supplier compliance & audit module (license approvals)

Phase 4: Worker Mobile App & Synchronization (Weeks 13-16)

- Lightweight mobile application build (React Native / Flutter)
- Availability calendar & accept/decline action flows
- Offline-first caching & delta sync engine
- End-to-end integration testing under simulated unstable network conditions

This specifications document incorporates every technical, operational, and regulatory constraint required for direct ingestion by a software engineering team or autonomous coding agent.

10. White-Label Architecture & Extensibility Standards

10.1 Design System & Dynamic Tokenization (Theming Engine)

- **Design Tokens Architecture:**
 - All user interfaces (Client Web App, Back-Office, Worker App) must consume dynamic design tokens managed via CSS variables, Tailwind configuration presets, or an abstracted theming layer (e.g., Theme UI / styled-tokens).
 - **Extensible Parameters:**
 - Color Palettes: Primary, secondary, surface, background, accent, and semantic alert tokens (Success, Warning, Timeout, Error).
 - Typography: Font families, scale, font weights, and letter spacing loaded via dynamic font injection.
 - Geometry: Border-radii, elevation/box-shadow scales, and button padding.
- **Asset Decoupling:**
 - Logos, favicons, brand illustrations, default empty-state artwork, and watermark badges must be fetched dynamically via an environment configuration file or asset storage bucket keyed by agency tenant ID.

10.2 Config-Driven Layouts & Feature Flagging

- **Module Registry:** Every major visual and functional block must be a standalone,

pluggable component governed by a single configuration manifest (`agency.config.ts / .json`).

- **Feature Toggles:**

TypeScript

// Example Agency Manifest Pattern

```
export const agencyConfig = {
  tenantId: "cuba-eco-travel",
  branding: {
    agencyName: "Authentic Cuba Expeditions",
    licenseNumber: "MINTUR-2026-XXXX", // Injected automatically into PDF
    footers
    primaryLocale: "es",
    supportedLocales: ["en", "fr", "es"],
  },
  theme: {
    palette: "ecoGreen", // References theme presets
    borderRadius: "md",
  },
  features: {
    interactiveSvgMap: true, // Enable/Disable province vector map
    culturalEventsBanner: true, // Enable/Disable event ticker
    recruitmentPortal: true, // Enable/Disable worker application
    form
    customItineraryBuilder: true, // Enable/Disable multi-stop builder
    instantBooking: false, // Toggle Request-to-Book vs Direct
    Checkout
    directBankTransferRail: true, // Enable/Disable SEPA/IBAN payment
    option
    creditCardGatewayRail: true, // Enable/Disable Card gateway (e.g.,
    TropiPay)
  },
  destinations: {
    geographyType: "cuba_provinces", // Can be swapped for another
    country/region map
  }
};
```

10.3 Internationalization (i18n) & Content Decoupling

- **Strict Content Isolation:** Zero marketing copy, destination descriptions, or static text inside UI component files.
- **Dictionary Layer:** All text strings must reside in isolated localization JSON files

(`/locales/{lang}.json`) or be driven dynamically via a headless CMS (e.g., Strapi, Directus, or Sanity).

- **PDF & Notification Templating:** Vouchers, invoices, contracts, and transactional email templates must be rendered from modular HTML/Handlebars templates populated by the active tenant config and localized message catalogs.

10.4 Extensible Deployment Strategy

- The repository must support:
 - **Tenant Forking:** A clear git branching strategy where updates to core backend routines (e.g., booking state machines, offline worker sync) can be upstreamed to forked client instances.
 - **Environment Parameterization:** Database connection strings, storage bucket URIs, gateway credentials, and map layers managed strictly through isolated `.env` configurations without codebase modifications.